RetailCore

Venture Viability Analysis

MIT World Peace University (MIT-WPU), Pune, Maharashtra

WADHWANI FOUNDATION

RetailCore

We provide a smart mobile POS and digital Udhaar credit ledger for neighborhood Kirana stores.



MIT World Peace University (MIT-WPU), Pune, Maharashtra



Context

Over 80% of India's 13M Kirana stores rely on paper ledgers.

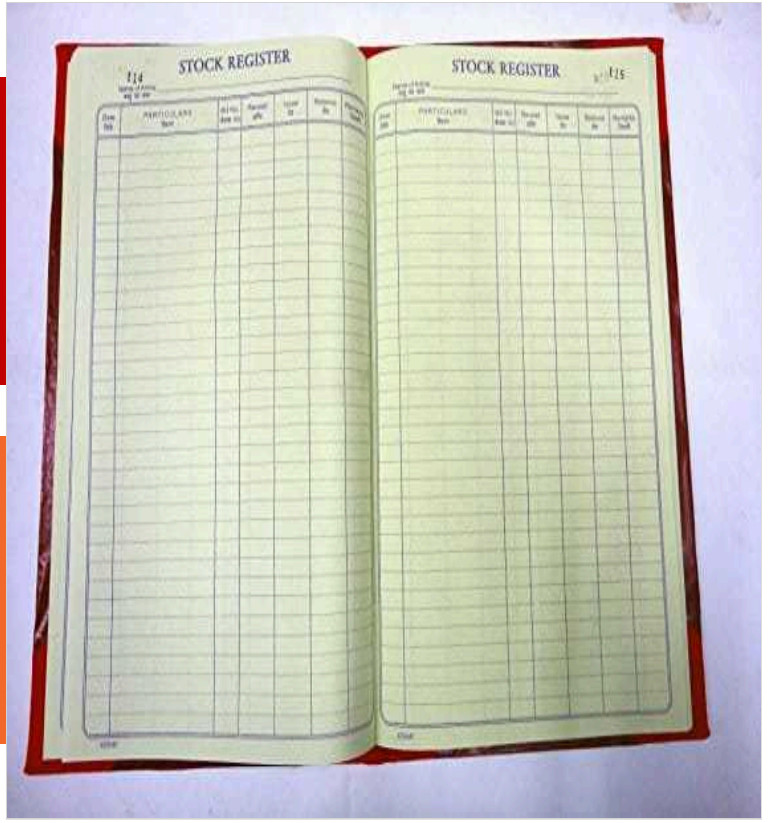
Problem Statement

Problem

80% of India's 13M Kirana stores use paper ledgers for sales and credit, causing debt losses and stock tracking errors.

Impact

Merchants lose 15% profit; consumers face billing disputes; FMCG distributors suffer delayed cash flows.



Problem Statement/Industry

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Problem Being Solved

Small-scale neighborhood retail merchants (Kirana stores and local shops) face severe operational inefficiencies, cash-flow bottlenecks, and inventory shrinkage caused by manual paper-based ledger tracking of informal customer credit (udhaar) and inaccurate stock recording.

Supporting Data

Over 80% of India's 13M Kirana stores rely on paper ledgers.

Source: EY Survey on Micro-Merchant. KPMG & FICCI Joint Report on Indian Retail and MSME Landscape.

Area
Business Services

Industry
Services

Domain
90% Unorganized Market: Over 90% of India’s retail sector spans 13 million Kirana stores, with over 80% relying strictly on

Problem Analysis

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Affected Stakeholders

Kirana Owners: 13M+ neighborhood retail shopkeepers running small grocery/daily-needs stores. Local Customers: Households using monthly informal credit (udhaar) for daily needs. FMCG Distributors: Wholesale suppliers delivering goods to micro-retailers on credit terms.

Impact on Stakeholders

Kirana Owners: Trap up to 20% working capital in bad credit and lose 15% profits to ledger errors. Local Customers: Face billing disputes and lack clear itemized proof for monthly credit balances. FMCG Distributors: Experience delayed collection cycles and erratic stock re-order patterns.

Root Causes

Existing ERP/POS software is too complex, expensive, and hardware-heavy for small shops. Paper ledgers require zero technical setup, making them a default despite errors and physical loss. Shopkeepers fear rigid digital debt collection will alienate loyal neighborhood customers.

Personal/Team Connect

As MIT-WPU tech students, we see local shopkeepers near our campus struggle with paper ledgers daily. We want to leverage our software skills to build a simple digital solution that brings financial clarity to micro-merchants, directly aligning with R.I.D.E. '26's "Insight to Impact" visionAs MIT-WPU tech students.

Target Customer Segments

Primary

Small neighborhood Kirana and daily-needs store owners



Secondary

LARGE SCALE RETAIL SHOPKEEPERS OR MINI MARKETS

Customer Segment & Persona

Primary Segment

Small neighborhood Kirana and daily-needs store owners

Secondary Segment

LARGE SCALE RETAIL SHOPKEEPERS OR MINI MARKETS

Persona

Ramesh Patil
Age in years: 42
Location: Metropolitan
Organizational Role: {Persona's primary role}
(if applicable)

Customer Profile

Education: High school pass out

Gender: Male

Occupation: Self-employed

Interests/Hobbies: Writing / Reading

Primary Source of Information: Traditional Media (TV, Newspapers)

Shopping Preference: Mostly Offline

Comfort with Technology: Medium

Favourite Social Media: WhatsApp (for daily communication)

Favourite Offline Gathering Spots: Pune Market Yard (APMC wholesale market)

Jobs-to-be-Done

Functional JTBD

Log daily cash and credit (udhaar) transactions instantly during peak store rush hours. Track outstanding customer credit balances and collect overdue payments on time without missing records. Monitor fast-moving versus slow-moving stock to prevent stockouts and shelf decay. Calculate daily and monthly net profits.

Emotional JTBD

Feel financially secure and in complete control of shop working capital and daily cash flow. Reduce anxiety and mental stress caused by misplaced paper ledgers and uncollected debts. Feel confident when reordering stock from wholesale suppliers without fear of hidden cash shortages.

Social JTBD

Maintain trust and strong neighborhood relationships with regular customers through clear, dispute-free billing. Politely request debt settlements without awkwardness or alienating loyal local families. Gain respect as a modern, successful shopkeeper among peer merchants and local trade associations.

Current Alternatives



Current Alternatives

Maintaining physical paper notebooks (khata books) to record daily customer credit entries. Using manual paper receipts and physical cash till drawers for daily accounting. Making personal phone calls or giving verbal reminders when customers visit the store to collect debts. Relying on visual shelf inspections and memory to reorder stock.



Gaps in Current Alternatives

Paper Ledgers: Highly vulnerable to physical damage, misplaced entries, billing errors, and lack automated reminder capabilities. Complex Desktop ERP/POS Systems: Overly expensive, hardware-heavy, complicated to learn, and too slow to operate during peak shop hours. Generic Digital Accounting Apps: Require too many tedious manual inputs per item.

Our Solution

Solution

RetailCore— a Kirana-store micro-ERP: smart POS, inventory, and digital Udhaar ledger for small Indian shops. Features: voice/barcode billing, GST calc, Cash/UPI/Udhaar checkout, 228+ products loaded

Core Technologies/ Methodologies

Fast barcode POS billing, pre-loaded 228+ FMCG items catalog, manual Udhaar credit ledger logging, and automated WhatsApp payment reminder trigger

Solution Design

Our Solution

RetailCore— a Kirana-store micro-ERP: smart POS, inventory, and digital Udhaar ledger for small Indian shops. Features: voice/barcode billing, GST calc, Cash/UPI/Udhaar checkout, 228+ products loaded

Key Features

Excludes multi-store inventory syncing, direct online payment gateway settlement, automated supplier purchase ordering, and advanced multi-year profit forecasting.

Uniqueness

Pre-loaded catalog of 228+ FMCG items for 1-click setup, automated WhatsApp Udhaar reminders, and a hardware-free mobile micro-ERP tailored for local Kiranas.

Solution Format:
A functional digital app prototype

Core Technologies/ Methodologies:
Fast barcode POS billing, pre-loaded 228+ FMCG items catalog, manual Udhaar credit ledger logging, and automated WhatsApp payment reminder trigger

Solution Benefits

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Functional Benefits

Save 60 minutes daily on manual bookkeeping, recover trapped working capital, eliminate credit accounting errors, and get real-time stock and daily profit visibility.

Emotional Benefits

Complete peace of mind over shop cash flow, zero stress about lost paper ledgers or forgotten debts, and confidence in managing daily working capital.

Social Benefits

Stronger neighborhood trust through clear itemized digital billing, polite dispute-free credit collection, and high respect as a modern, digitized store owner.

Macro Benefits

Empowers unorganized Kirana merchants with financial clarity, strengthens local MSME liquidity, speeds digital inclusion, and eliminates paper ledger waste.

Competitors



Direct

Khatabook, Vyapar, MyBillBook, OkCredit, and Petpooja Invoice.
Loyverse POS, Kyte, Square for Retail, Lightspeed, and POS-Blox

Indirect

Paytm Business, PhonePe Soundbox, TallyPrime, Udaan, and Jumbotail.
Shopify POS, QuickBooks, Clover, Toast, and Zoho Books Global

Our UVP

Hardware-free Kirana mobile POS with 228+ pre-loaded FMCG items and WhatsApp Udhaar reminders.

Competitors

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Direct Competitors

Khatabook, Vyapar, MyBillBook, OkCredit, and Petpooja Invoice.

Indirect Competitors

Paytm Business, PhonePe Soundbox, TallyPrime, Udaan, and Jumbotail.

Direct Competitors Globally

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Indirect Competitors Globally

Shopify POS, QuickBooks, Clover, Toast, and Zoho Books Global

Macro Analysis

Favourable Trends

AREA	DESCRIPTION
Technology	Universal QR/UPI adoption makes Kiranas comfortable using smartphone financial apps daily.
Technology	Govt initiatives and ONDC drive Tier-2/3 Kirana owners to adopt digital store tools.

Unfavourable Trends

AREA	DESCRIPTION
Other	Rapid 10-minute delivery apps (Zepto, Blinkit) reduce neighborhood Kirana store footfall and sales
Technology	Small shop owners heavily resist paying recurring software fees, leading to high free-tier churn.

Data Sources:

Bain & Company & EY India: India Retail & FMCG Report (insights on paper ledger usage, credit loss ratios, and micro-merchant digital adoption barriers).

Back-of-the-Envelope Financial Projections

Currency: Indian Rupee (INR)

Chosen Business Model: Product Manufacturing

AREA	YEAR 1	YEAR 2	YEAR 3
Revenues	1200000	3600000	9000000
Total Expenses	960000	1920000	3456000
Profit	240000	1680000	5544000

Prototype

Prototype Format

The prototype format is digital, presented as an interactive user interface flow. It will be built u

Functionality included in the Prototype

Presentation Method:The prototype will be built as an interactive user interface flow in Figma. It will features clickable hotspots, allowing users to navigate between screens just like a live applica

Functionality NOT included in the Prototype

The prototype will exclude fully functional back-end databases, live third-party integrations, and automated notifications. These complex systems require backend programming and are non-essential for

{Upload another picture}

Revenue Models / Pricing

Revenue Model (Primary)

Saas (Software as a Service) Model

Revenue Model (Secondary)

Saas (Software as a Service) Model

Lean Canvas

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Problem
80% of India's 13M Kirana stores use paper ledgers for sales and credit, causing debt losses and stock tracking errors.

Solution
RetailCore— a Kirana-store micro-ERP: smart POS, inventory, and digital Udhaar ledger for small Indian shops. Features: voice/barcode billing, GST calc, Cash/UPI/Udhaar checkout, 228+ products loaded

Key Metric
Active Usage: Daily/Monthly Active Merchants (DAM/MAM). Transaction Volume: Gross Merchandise Value (GMV) billed on app. Udhaar Efficiency: Debt recovery rate via WhatsApp. Retention: Monthly subsc

Unique value Proposition
Hardware-free Kirana mobile POS with 228+ pre-loaded FMCG items and WhatsApp Udhaar reminders.

Unfair Advantage
Proprietary pre-configured database of 228+ regional FMCG products for zero-typing setup, paired with deep hyper-local ground insights from direct field engagement with Tier-2/3 Kirana owners.

Channels
Direct Digital: Google Play Store downloads. Field Sales: Feet-on-street team for direct door-to-door shop onboarding. Local Networks: Merchant associations and referral programs.

Customer Segment
Small neighborhood Kirana and daily-needs store owners,LARGE SCALE RETAIL SHOPKEEPERS OR MINI MARKETS

Cost Structure
Cloud Infrastructure: AWS/Firebase server hosting. WhatsApp API: Webhook messaging credits for Udhaar alerts. Field Sales: Onboarding representative stipends. Software Maintenance: Ongoing app R&D.

Revenue Streams
Saas (Software as a Service) Model, Saas (Software as a Service) Model

Financials

Revenue Models/Streams

- Saas (Software as a Service) Model
- Saas (Software as a Service) Model

Pricing

- **Unit of Sale:** Monthly app subscription
- **Selling price per unit:** ₹100 per month

First Year Projections

Revenues: 3600000

Operating Profits: ₹15,80,000

Revenue Models / Pricing

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Revenue Model (Primary)

Saas (Software as a Service) Model

Unit of Sale	Sale Price per Unit	Expected units to be sold in Year 1	Expected growth in monthly sales
Monthly app subscription	₹100 per month	50 subscriptions	12% per month

Costs & Revenues: Key Assumptions

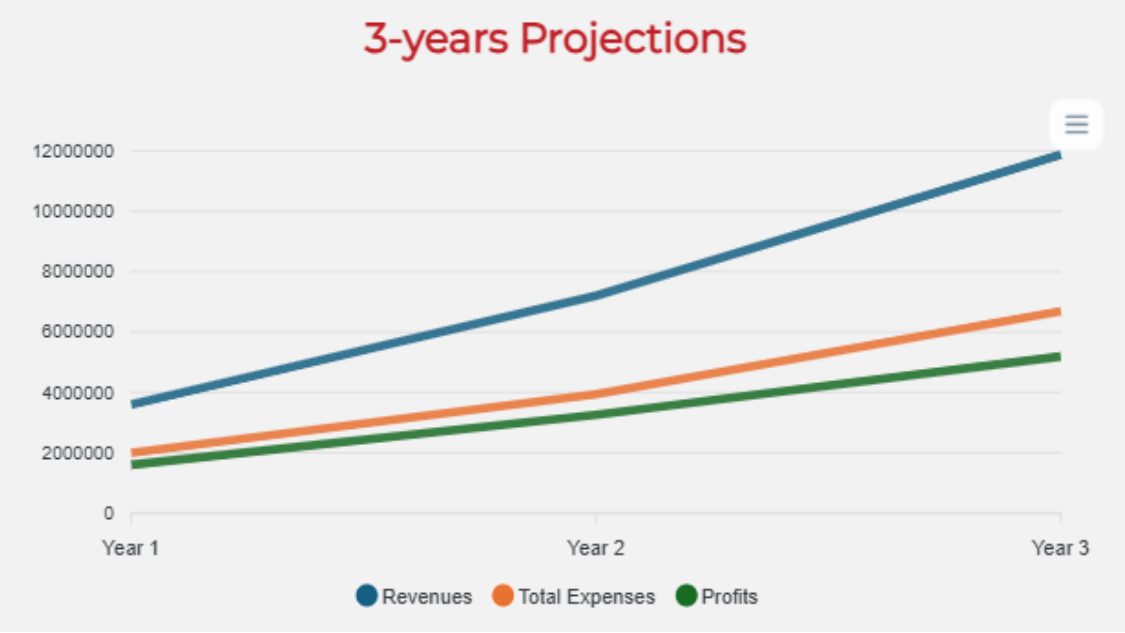
		Year 1 Assumptions	Going from Year 1 to Year 2		Going from Year 2 to Year 3	
TYPE	DESSCRIPTION	Financials, Absolute	Business Assumptions	Financial Decisions, % change	Business Assumptions	Financial Decisions, % change
SETUP COSTS	App Architecture & Infrastru	200000	1. Core app framework completed; no duplicate setup required. 2. Double monthly shop onboarding across 2 new Tier-2 city hubs. 3. Negotiate bulk WhatsApp API volume discounts (-10% cost). 4. Double field team and cloud server capacity to support growth.	100	1. Scale cloud architecture across new regional servers. 2. Grow active store onboardings by 80% through referral networks. 3. Maintain bulk API rates at higher transactional scale. 4. Expand engineering and dedicated regional support team by 80%	0
SALES	Annual Subscription Price p	1200		0		0
	# Stores Onboarded per Mc	250		100		65
VARIABLE COSTS	WhatsApp API & Usage Co	100		-10		0
FIXED COSTS (MONTHLY)	Field Onboarding Rep Stipe	60000		100		80
	Marketing	20000		100		80
	Server Hosting & Cloud DB	25000		100		80
	Customer Support & Mainte	20000		100		80

Profit & Loss Projections: Summary

	Year 1	Year 2	Year 3
Revenues	3600000	7200000	11880000
Total Expenses	2000000	3940000	6691000
Profits	1600000	3260000	5189000

Performance & Break-Even Analysis

Year 1 Revenues	Gross Profits for Year 1
3600000	₹32,80,000
Net Profits for Year 1	**Break-even Month**
₹15,80,000	Month 5



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Thank You!

